永林电子股份有限公司
Y.LIN ELECTRONICS CO.,LTD.

Data Sheet

Customer: A01

Part No: YLTFT-HM144-101

Sample No:

Description: TFT LCD Module,Transmissive type,Normally White. LCD Module: Graphic 128RGB*128Dot-matrix

Item No:

Long Code:

Customer			
Check	Inspection	Approval	Date

Y.LIN				
Drawn	Check	Approval	CSO Approval	Date
				2026.08.25

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Revision History

Date	Rev.	Description of changes
2026.08.25	V1.0	Initial Release

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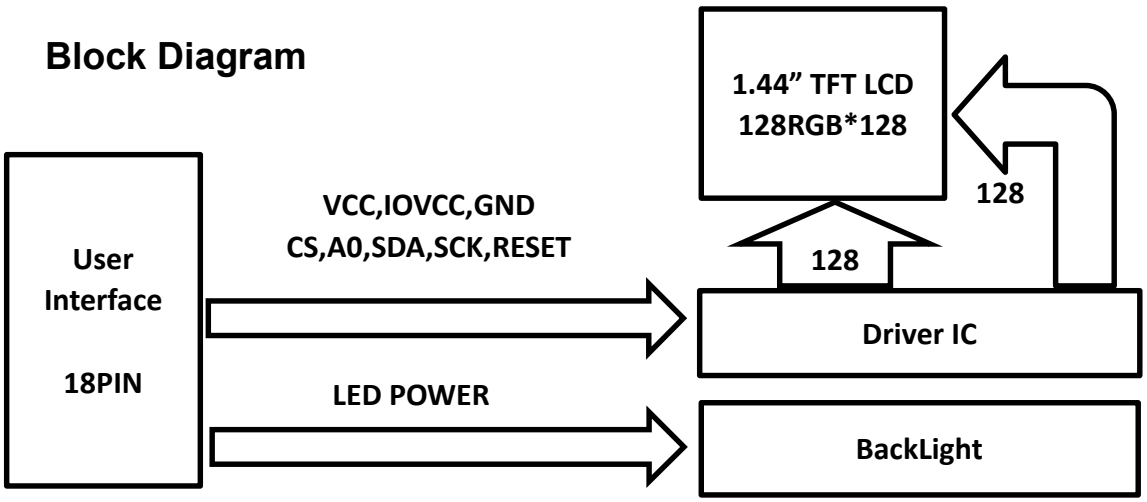
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1.0 GENERAL DESCRIPTION

1.1 Introduction

HM144-101 is a color active matrix TFT LCD module using amorphous silicon TFT's (Thin Film Transistors) as an active switching devices. It is a transmissive type display operating in the normal white. The TFT-LCD has a 1.44 inch diagonally measured active area with resolutions (128 horizontal by 128 vertical pixel arrays). Each pixel is divided into RED, GREEN, BLUE dots which are arranged in vertical stripe and this panel can display 262K colors.

Block Diagram



1.2 General Specification

Parameter	Specification	Unit
Active area	25.5(H) × 26.5(V)	mm
Module outline	33.3(H) × 42.94(V) × 2.2(T)	mm
Number of pixels	128(H) ×RGB ×128(V)	pixels
Pixel pitch	0.1992(H) ×0.207(V)	mm
Display colors	65K/262K	colors
Color gamut	53typ. NTSC	%
Display mode	Normally White	/
Viewing direction (human eye)	U/D/L/R Typ. 55/50/60/60 Min,45/40/50/50	Deg
Driver IC	ST7735P	/
Backlight Type	2LEDS / White	/
Interface Type	4SPI	/

2.0 ABSOLUTE MAXIMUM RATINGS

The followings are maximum values which, if exceed, may cause faulty operation or damage to the unit.

Parameter	Symbol	Min	Max	Unit	Remarks
Power for Circuit Driving	VCC	-0.3	4.6	V	Ta=25+/-2°C
Power for Circuit Logic	IOVCC	-0.3	4.6	V	
Operating temperature	Top	-20	70	℃	
Storage temperature	Tst	-30	80	℃	
Operating Ambient Humidity	Hop	10	90	%RH	At 60°C
Storage Humidity	Hst	10	90	%RH	At 60°C

3.0 ELECTRICAL SPECIFICATIONS

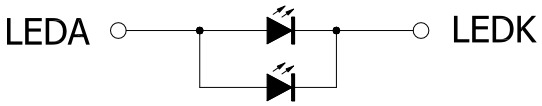
3.1 Electrical Specifications

Parameter	Symbol	Min	Typ	Max	Unit
Power for analog	Vcc -GND	2.65	2.8	3.3	V
I/O power supply	IOVCC	1.65	1.8	3.3	V
Input Current	Idd	-	10	30	mA
Input voltage ' H ' level	Vih	0.7IOVCC	--	IOVCC	℃
Input voltage ' L ' level	Vil	GND	0	0.3IOVCC	
Output voltage ' H ' level	Voh	0.8IOVCC	--	IOVCC	℃
Output voltage ' L ' level	Vol	GND	0	0.2IOVCC	RH

3.2 Backlight Characteristics

Item	Symbol	Min	Typ	Max	Unit	Remark
Forward voltage	Vf	2.8	3.0	3.2	V	2 strings
Supply current of LED	Per string	-	15	20	mA	1LED
Total Supply current of LED	I_{LED}Total	-	30	40	mA	2 strings

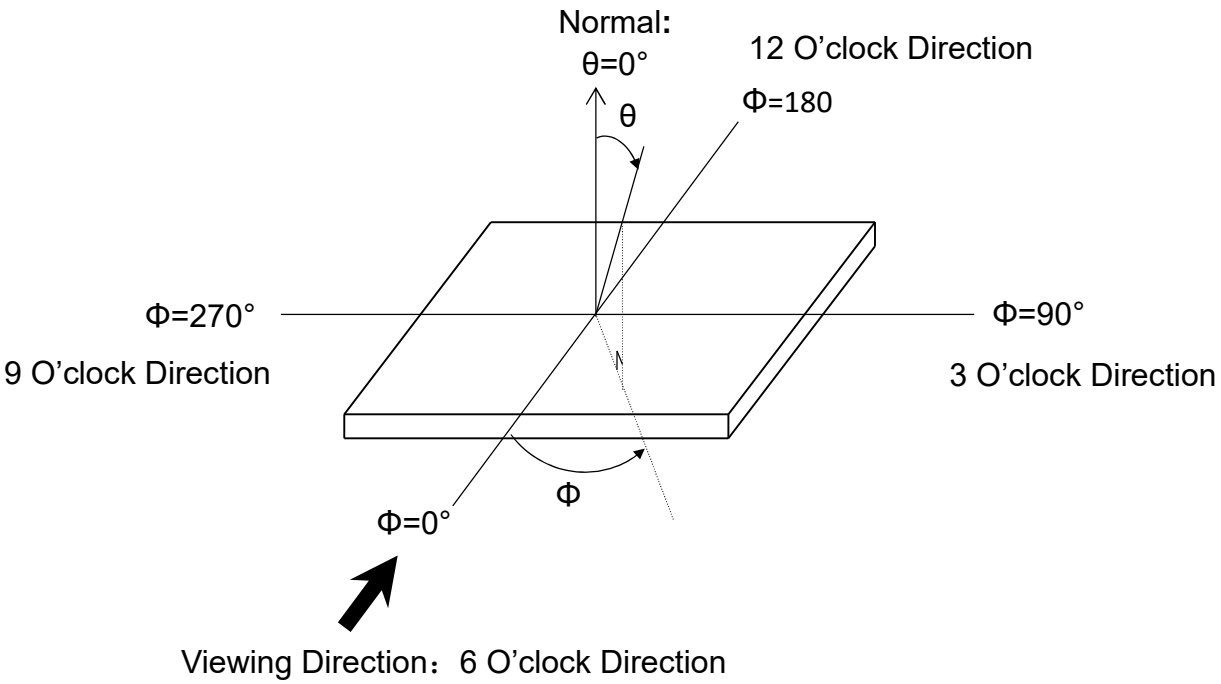
Backlight Circuit diagram



4.0 OPTICAL SPECIFICATIONS

Item		Symbol	Condition	Min	Typ	Max	Unit	Note
Response time		Tr+Tf	θ=0° Φ=0° Ta=25℃	-	25	40	ms	/
Contrast ratio		CR		400	500	-	-	/
Luminance (White)		LV		-	220	-	nit	/
Luminance uniformity		δ WHITE		70	80	-	%	/
Viewing angle range		Θ3	CR>10	50	60	-	deg	/
		Θ9		50	60	-	deg	
		Θ6		40	50	-	deg	
		Θ12		45	55	-	deg	
CIE(x,y) chromaticity	Red	x	θ=0° Φ=0° Ta=25℃	Typ-0.015	0.612	Typ+0.015	/	/
		y			0.328			
	Green	x			0.295			
		y			0.571			
	Blue	x			0.141			
		y			0.159			
	White	x			0.301			
		y			0.338			

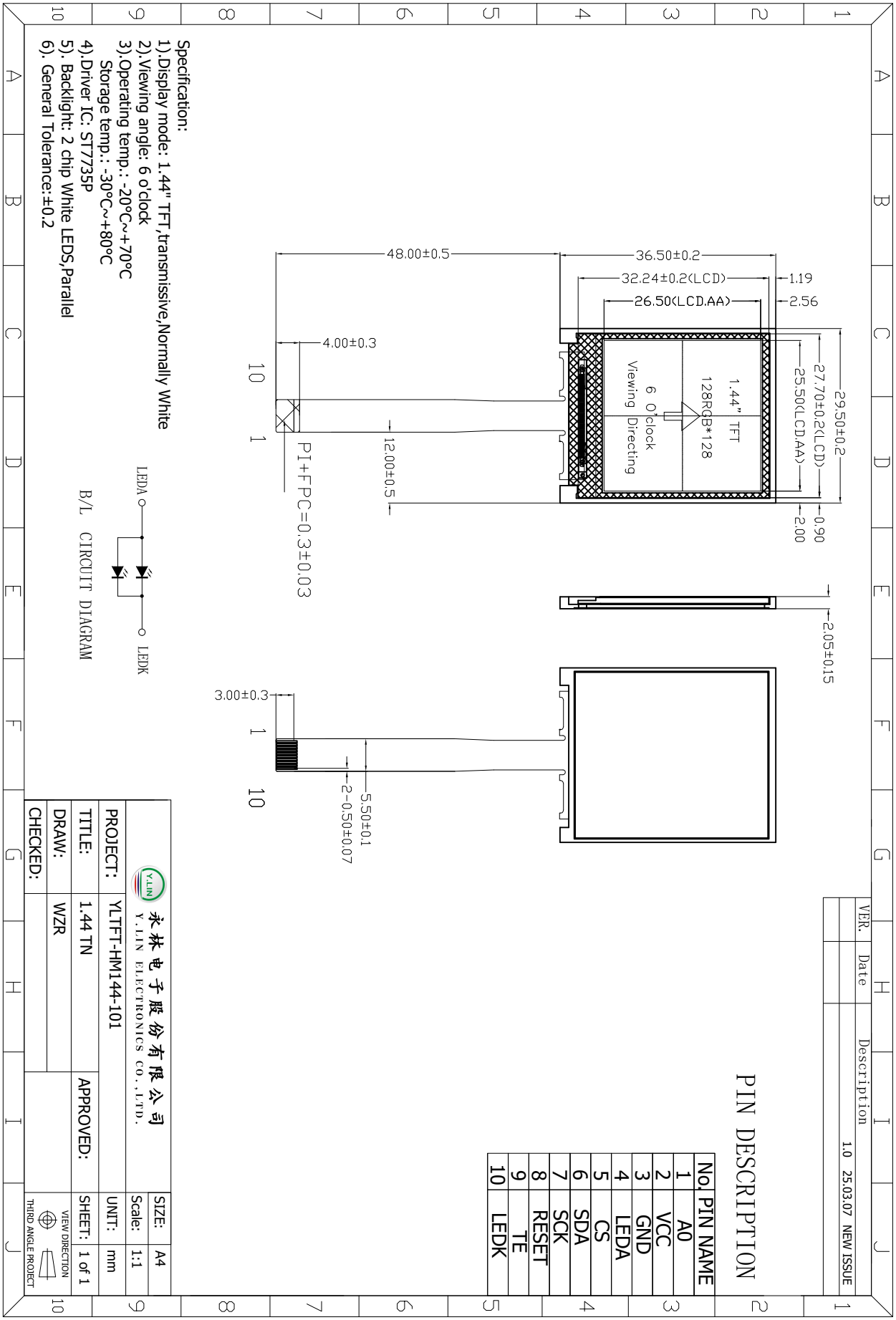
Definition of Viewing Angle θ and Φ



5.0 PIN DESCRIPTION

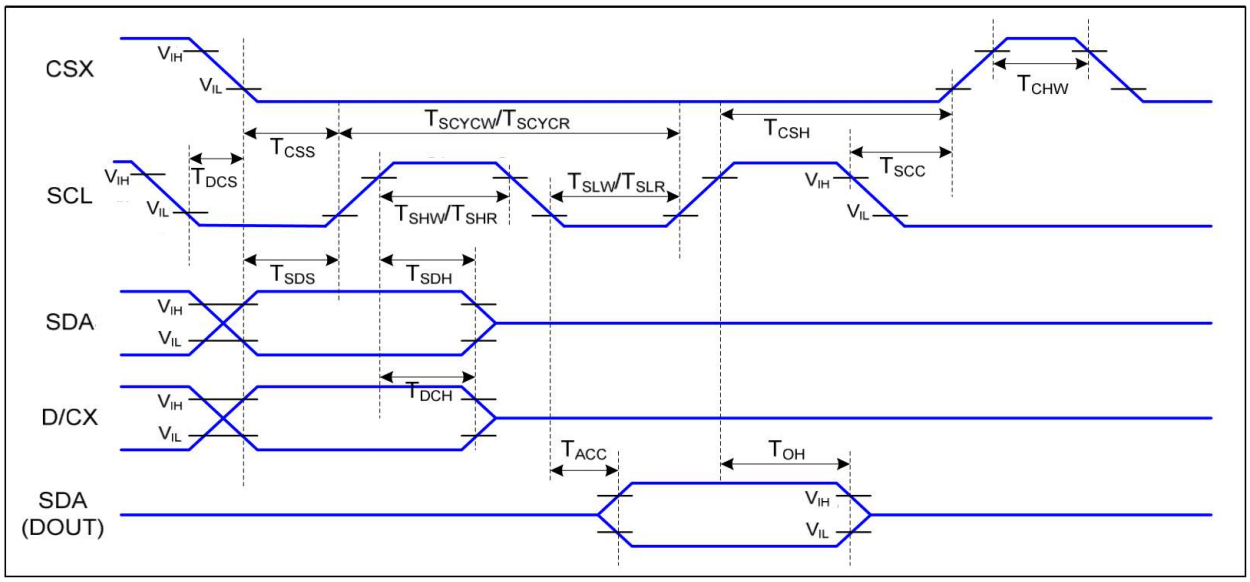
Pin.No	Symbol	Description
1	A0	Display data/command selection pin;
2	VCC	PowerSupplyforAnalog,Digital System and Booster Circuit
3	GND	Power Ground
4	LEDA	LEDA node
5	CS	Chip select input pin. Enabled when CS is “L”.
6	SDA	Serial data input
7	SCL	Serial clock input
8	RESET	Areset pin
9	TE	Tearing effect output pin to synchronize MPU to frame writing
10	LEDK	LED Cathode

6.0 Mechanical Characteristics



7.0 TIMING CHARACTERISTICS

Parallel Interface Timing Characteristics (4-wire serial Interface)



Note: Ta = -30 to 70 ° C, IOVCC=1.65V to 3.3V, VCC=2.5V to 3.3V, GND=0V

Signal	Symbol	Parameter	MIN	MAX	Unit	Description
CSX	TCSS	Chip Select Setup Time (Write)	45		ns	
	TCSH	Chip Select Hold Time (Write)	45		ns	
	TCSS	Chip Select Setup Time (Read)	60		ns	
	TSCC	Chip Select Hold Time (Read)	65		ns	
	TCHW	Chip Select “H” Pulse Width	40		ns	
SCL	TSCYCW	Serial Clock Cycle (Write)	66		ns	-Write Command & Data Ram
	TSHW	SCL “H” Pulse Width (Write)	15		ns	
	TSLW	SCL “L” Pulse Width (Write)	15		ns	
	TSCYCR	Serial Clock Cycle (Read)	150		ns	-Read Command & Data Ram
	TSHR	SCL “H” Pulse Width (Read)	60		ns	
	TSLR	SCL “L” Pulse Width (Read)	60		ns	
D/CX	TDCS	D/CX Setup Time	10		ns	
	TDCH	D/CX Hold Time	10		ns	
SDA (DIN) (DOUT)	TSDS	Data Setup Time	10		ns	For Maximum CL=30pF For Minimum CL=8pF
	TSDH	Data Hold Time	10		ns	
	TACC	Access Time	10	50	ns	
	TOH	Output Disable Time	15	50	ns	

8.0 RELIABILITY

N0.	TEST ITEM	CONDITIONS
1	High Temperature Storage	80℃；96 hrs
2	Low Temperature Storage	-30℃；96 hrs
3	HighTemperature Operation	70℃；96 hrs
4	Low Temperature Operation	-20℃；96 hrs
5	High Temperature and High Humidity Operation	60℃, 90% RH；96 hrs
6	Thermal shock(Storage)	-20℃(0.5Hr)→70℃(0.5Hr) 100 Cycles

NOTE:

1. All judgement of display are performed after temperature of panel return to room temperature.
2. Display function should be no change under normal operating condition.
3. Under no condensation of dew.
4. WE only guarantee the above 6 test items, and without guarantee the others.

9.0 PRECAUTIONS

9.1 Handing Precautions

(1) The display panel is made of glass and polarizer. As glass is fragile, it tends to become or chipped during handling especially on the edges. Please avoid dropping or jarring. Do not subject it to a mechanical shock by dropping it or impact.
(2) If the display panel is damaged and the liquid crystal substance leaks out, be sure not to get any in your mouth. If the substance contacts your skin or clothes, wash it off using soap and water.
(3) Do not apply excessive force to the display surface or the adjoining areas since this may cause the color tone to vary. Do not touch the display with bare hands. This will stain the display area and degraded insulation between terminals (some cosmetics are determined to the polarizer).
(4) The polarizer covering the display surface of the LCD module is soft and easily scratched. Handle this polarizer carefully. Do not touch, push or rub the exposed polarizers with anything harder than an HB pencil lead (glass, tweezers, etc.). Do not put or attach anything on the display area to avoid leaving marks on. Condensation on the surface and contact with terminals due to cold will damage, stain or dirty the polarizer. After products are tested at low temperature they must be warmed up in a container before coming is contacting with room temperature air.
(5) If the display surface becomes contaminated, breathe on the surface and gently wipe it with a soft dry cloth. If it is heavily contaminated, moisten cloth with one of the following solvents
- Isopropyl alcohol
- Ethyl alcohol

Do not scrub hard to avoid damaging the display surface.
(6) Solvents other than those above-mentioned may damage the polarizer. Especially, do not use the following.
- Water
- Ketone
- Aromatic solvents

Wipe off saliva or water drops immediately, contact with water over a long period of time may cause deformation or color fading. Avoid contacting oil and fats.
(7) Exercise care to minimize corrosion of the electrode. Corrosion of the electrodes is accelerated by water droplets, moisture condensation or a current flow in a high-humidity environment.
(8) Install the LCD Module by using the mounting holes. When mounting the LCD module make sure it is free of twisting, warping and distortion. In particular, do not forcibly pull or bend the I/O cable or the backlight cable.
(9) Do not attempt to disassemble or process the LCD module.
(10) NC terminal should be open. Do not connect anything.
(11) If the logic circuit power is off, do not apply the input signals.
(12) Since LCM has been assembled and adjusted with a high degree of precision, avoid applying excessive shocks to the module or making any alterations or modifications to it.
- Do not alter, modify or change the shape of the tab on the metal frame.
- Do not make extra holes on the printed circuit board, modify its shape or change the positions of components to be attached.
- Do not damage or modify the pattern writing on the printed circuit board.
- Absolutely do not modify the zebra rubber strip (conductive rubber) or heat seal connector.
- Except for soldering the interface, do not make any alterations or modifications with a soldering iron.
- Do not drop, bend or twist LCM.

9.2 Storage Precautions

When storing the LCD modules, the following precaution is necessary.
(1) Store them in a sealed polyethylene bag. If properly sealed, there is no need for the dessicant.
(2) Store them in a dark place. Do not expose to sunlight or fluorescent light, keep the temperature between 0° C and 35° C.
(3) The polarizer surface should not come in contact with any other objects. (We advise you to store them in the container in which they were shipped).

9.3 Others

Liquid crystals solidify under low temperature (below the storage temperature range) leading to defective orientation or the generation of air bubbles (black or white). Air bubbles may also be generated if the module is subject to a low temperature.
If the LCD modules have been operating for a long time showing the same display patterns, the display patterns may remain on the screen as ghost images and a slight contrast irregularity may also appear. A normal operating status can be regained by suspending use for some time. It should be noted that this phenomenon does not adversely affect performance reliability.
To minimize the performance degradation of the LCD modules resulting from destruction caused by static electricity etc., exercise care to avoid holding the following sections when handling the modules.
- Exposed area of the printed circuit board.
-Terminal electrode sections.